Stony Brook Institute at Anhui University

安徽大学纽约石溪学院

Chapter 2 Random variable and their probability distributions

Section 2.3 Continuous random variable and its probability density

- Probability density and its properties
- Cumulative distribution function of continuous random variable
- The distribution of common continuous random variables
- Examples

Probability density function(pdf)

Definition Let X be an r.v., if there exists a nonnegative and integrable function $f(x)$ which is defined on $\mathbb{R}^1$, and $f(x)$ satisfy

$$F(x) = \int_{-\infty}^{x} f(t)\mathrm{d}t$$

where $F(x)$ is cdf of X. Then r.v. X is called a continuous random variable, and $f(x)$ is called the probability density function (pdf for short) of X.

In calculus we know that $\int_{-\infty}^{x} f(t)\mathrm{d}t$ is a differential function, so $F(x)$ is continuous.

(1). $f(x) \geq 0$ **(2).** $\int_{-\infty}^{+\infty} f(x)dx = 1$

Probability density function(pdf)

Properties

Example： Prove that：$f(x)=\frac{1}{2}e^{-|x|}\quad(-\infty<x<+\infty)$ is a pdf of a continuous r.v.

Answer：

（1）Absolutely，$f(x)>0\quad(-\infty<x<\infty)$

(2) and $\int_{-\infty}^{+\infty} f(x)dx=\int_{-\infty}^{+\infty}\frac{1}{2}e^{-|x|}dx$

$$=\frac{1}{2}\int_{-\infty}^{0}e^{x}dx+\frac{1}{2}\int_{0}^{+\infty}e^{-x}dx$$

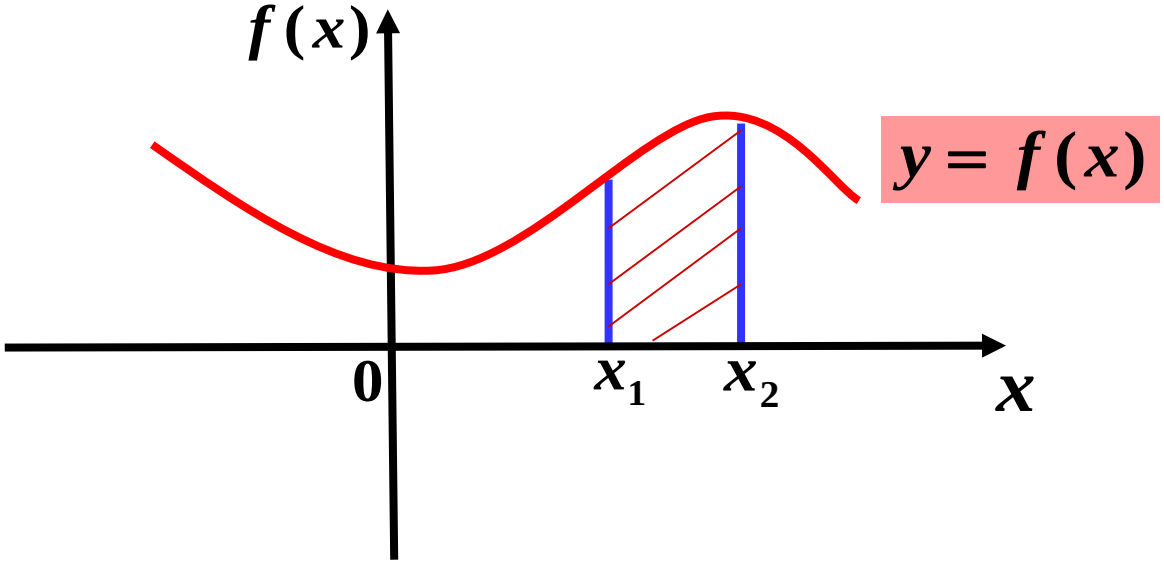
$$=\frac{1}{2}+\frac{1}{2}=1$$

So，$f(x)=\frac{1}{2}e^{-|x|}\quad(-\infty<x<+\infty)$ is a pdf of a continuous r.v.

Probability density function(pdf)

Properties

(3) $P\{x_1 < X \le x_2\} = F(x_2) - F(x_1) = \int_{x_1}^{x_2} f(x)dx$



$$P\{X \le a\} = F(a) = \int_{-\infty}^{a} f(x)dx$$

$$P\{X > a\} = 1 - F(a) = \int_{a}^{+\infty} f(x)dx$$

Probability density function(pdf)

(4) If function f(x) is continuous at x, then $F'(x) = f(x)$.

$$P(x < X \le x + \Delta x) = F(x + \Delta x) - F(x) = \int_x^{x+\Delta x} f(t)\,dt$$

$$\approx f(x)\Delta x$$

- This means that the probability of falling on interval $(x, x+\Delta x]$ is approximately equal to $f(x)\Delta x$, and $f(x)\Delta x$ is a probability differential.
- The value of $f(x)$ directly affects the probability, so $f(x)$ does describe the probability distribution of continuous random variables.

Probability density function(pdf)

Example. Suppose cdf of r.v.X is $F(x)=\begin{cases} 0, & x<0 \\ x^2, & 0\le x\le 1 \\ 1, & x>1 \end{cases}$,

Calculate: (1) the probability of X in (0.3,0.7); (2) the pdf.of X.

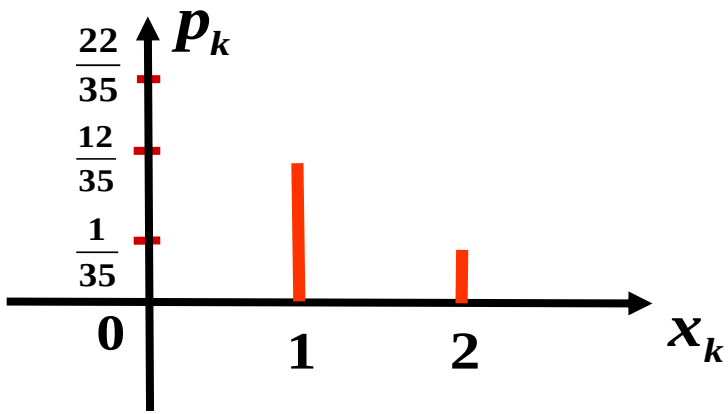
Answer: **(1)** $P(0.3<X<0.7)=F(0.7)-F(0.3)=0.4$

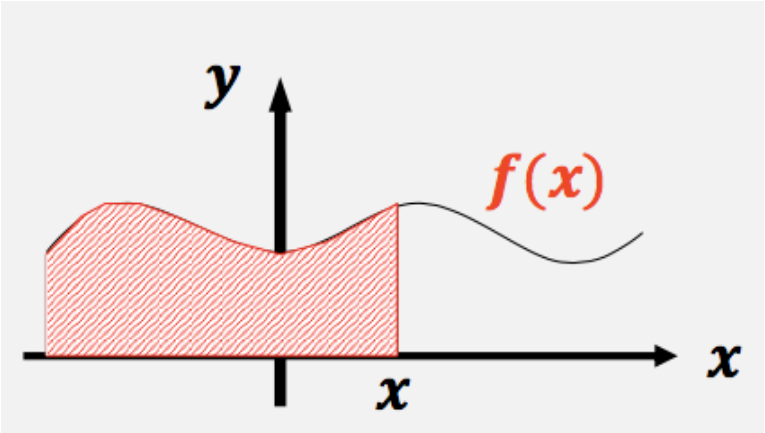
(2) $f(x)=\dfrac{dF(x)}{dx}=\begin{cases} 2x & 0\le x<1 \\ 0 & \text{else} \end{cases}$

- **Noticed** that the derivative of F(x) does not exist at x = 1.
- According to the property that changing the value of the integrand function at individual points does not affect the integration result, the value of $F'(x)$ can be arbitrarily specified at the meaningless point.

Probability density function(pdf)

The difference between continuous random variable and discrete random variable:





discrete r.v: $P(X = x_k) \geq 0$

continuous r.v: $P(X = x_k) = 0$

Probability density function(pdf)

Prove: For continuous r.v: $P(X = x_k) = 0$

Suppose $\forall\, x_0 \in (-\infty, \infty)$, increment by Δx.

$$\because\ 0 \le P(X = x_0) \le P(x_0 < X \le x_0 + \Delta x)$$
$$= F(x_0 + \Delta x) - F(x_0)$$
$$= \int_{x_0}^{x_0+\Delta x} f(t)dt$$

$\Delta x \to 0$

$$0 \le P(X = x_0) \le \lim_{\Delta x \to 0} \int_{x_0}^{x_0+\Delta x} f(t)dt = 0$$

$$\Rightarrow \quad P(X = x_k) = 0$$

Probability density function(pdf)

▲ $P(X = x_k) = 0$

$$P(x_1 \le X < x_2) = P(x_1 < X \le x_2)$$

$$= P(x_1 \le X \le x_2)$$

$$= P(x_1 < X < x_2)$$

$$= \int_{x_1}^{x_2} f(x)dx = F(x_2) - F(x_1)$$

Probability density function(pdf)

▲ For a ***continuous / discrete r.v*** **X**

$$\mathbf{X} = \emptyset \Leftrightarrow \mathbf{P}\{\mathbf{X}\} = \mathbf{0}$$

$$\mathbf{X} = \mathbf{S} \Leftrightarrow \mathbf{P}\{\mathbf{X}\} = \mathbf{1}$$

- Impossible event ?
- Inevitable event ??

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CDF(c.d.f) for a continuous r.v X

$$F(x)=\int_{-\infty}^{x} f(x)dx$$

- F (x) determines the distribution function $f(x)$;
- $f(x)$ is the derivative of F (x);
- and $f(x)$ is an original function of F (x).

It can be verified that $F(x)$ has the **properties** :

(1) $F(x)$ is a monotone non decreasing function;

(2) $0 \le F(x) \le 1$;

(3) $F(x)$ is a right continuous function.

CDF(c.d.f) for a continuous r.v X

Example

Suppose function $F(x)=\begin{cases}\sin x & 0\le x\le \pi \\ 0 & \text{else}\end{cases}$,

Can $F(x)$ be the distribution function of a continuous r.v.?

Answer:

(1) F(x) falls on $[\pi/2,\pi]$;

(2) $F(+\infty)=\lim\limits_{x\to+\infty}F(x)=0$

Thus, $F(x)$ can't be the distribution function of a continuous random variable .

CDF(c.d.f) for a continuous r.v X

Example 2.3.1: Let the PDF of r.v. X be

$$f(x)=\begin{cases} kx, 0\le x<3 \\ 2-\frac{x}{2}, 3\le x\le 4 \\ 0, the\ rest\ of\ x \end{cases}$$

(1) Find k; (2)CDF of X; (3) $P\{1<x\le\frac{7}{2}\}$

Solution:

(1) k =1/6 ; (2)

$$F(x)=\begin{cases} 0, & x<0, \\ \int_0^x \frac{x}{6}dx, & 0\le x<3, \\ \int_0^3 \frac{x}{6}dx+\int_3^x (2-\frac{x}{2})dx, & 3\le x<4, \\ 1, & x\ge 4. \end{cases}$$

(3) $P\{1<X\le\frac{7}{2}\}=F(\frac{7}{2})-F(1)=\frac{41}{48}.$

CDF(c.d.f) for a continuous r.v X

Example

Let $f(x)=\begin{cases}\dfrac{x}{a}e^{-\frac{x^2}{2a}} & x\geq 0\\ 0 & x<0\end{cases}$ $(a>0)$

Find (1) cdf. X; (2) $P(0\leq X<1)$

Answer:

(1) $F(x)=\int_{-\infty}^{x} f(t)dt$

when $x<0$, $f(x)=0$

$$F(x)=\int_{-\infty}^{x} 0\cdot dt=0$$

CDF(c.d.f) for a continuous r.v X

$$f(x)=\begin{cases}\dfrac{x}{a}e^{-\frac{x^2}{2a}} & x\ge 0\\ 0 & x<0\end{cases}$$

when $x \ge 0$, $f(x)=\dfrac{x}{a}e^{-\frac{x^2}{2a}}$

$$F(x)=\int_{-\infty}^{0} 0\cdot dt + \int_{0}^{x}\frac{t}{a}e^{-\frac{t^2}{2a}}dt$$

$$=[-e^{-\frac{t^2}{2a}}]\Big|_0^x = 1-e^{-\frac{x^2}{2a}}$$

$$F(x)=\begin{cases}0 & x<0\\ 1-e^{-\frac{x^2}{2a}} & x\ge 0\end{cases}$$

(2) $P(0\le X<1) = F(1)-F(0) = 1-e^{-\frac{1}{2a}}$

CDF(c.d.f) for a continuous r.v X

Example. Suppose $f(x)$ is the pdf of continuous *r.v.* X, and

$$f(x)=\begin{cases}\dfrac{2}{\pi}\sqrt{1-x^2}, & -1\le x\le 1\\ 0, & \text{else}\end{cases}$$

Find : $F(x)$.

Answer:

$$F(x)=P(X\le x)=\int_{-\infty}^{x} f(t)dt$$

(1) when $x<-1$, $F(x)=0$

(2) when $-1\le x<1$, $F(x)=\int_{-\infty}^{-1} 0\cdot dt+\int_{-1}^{x}\frac{2}{\pi}\sqrt{1-t^2}dt$

$$=\frac{x}{\pi}\sqrt{1-x^2}+\frac{1}{\pi}\arcsin x+\frac{1}{2}$$

CDF(c.d.f) for a continuous r.v X

(3) When $x \geq 1$,

$$F(x) = \int_{-\infty}^{-1} 0 \cdot dt + \int_{-1}^{1} \frac{2}{\pi}\sqrt{1-t^2}\,dt + \int_{1}^{x} 0 \cdot dt = 1$$

$$F(x) = \begin{cases} 0, & x < -1 \\ \dfrac{x}{\pi}\sqrt{1-x^2} + \dfrac{1}{\pi}\arcsin x + \dfrac{1}{2}, & -1 \leq x < 1 \\ 1, & x \geq 1 \end{cases}$$

CDF(c.d.f) for a continuous r.v X

Example: The total running time (unit: hour) of an electronic computer before destruction is a continuous random variable, and its density function is:

$$f(x)=\begin{cases}\lambda e^{-\frac{x}{100}} & x\geq 0\\ 0 & x<0\end{cases}$$

Calculate:

(1) the value of λ;

(2) The probability that the computer will run for 50 to 150 hours before destruction;

(3) Probability of running time less than 100 hours.

CDF(c.d.f) for a continuous r.v X

Answer:

(1) $$\because\ 1=\int_{-\infty}^{+\infty} f(x)dx = \int_{0}^{+\infty} \lambda e^{-\frac{x}{100}}dx$$

$$= -100\lambda e^{-\frac{x}{100}}\Big|_{0}^{+\infty} = 100\lambda$$

$$\therefore\quad \lambda = \frac{1}{100}$$

(2) $$P(50 < X < 150) = \int_{50}^{150} \frac{1}{100} e^{-\frac{x}{100}} dx$$

$$f(x)=\begin{cases} \lambda e^{-\frac{x}{100}} & x \geq 0 \\ 0 & x < 0 \end{cases}$$

$$= -e^{-\frac{x}{100}}\Big|_{50}^{150} = 0.384$$

CDF(c.d.f) for a continuous r.v X

$$f(x)=\begin{cases}\lambda e^{-\frac{x}{100}} & x\geq 0\\ 0 & x<0\end{cases}$$

(3) $P(X<100) = \int_0^{100} \frac{1}{100} e^{-\frac{x}{100}} dx$

$$= 1-e^{-1} \approx 0.633$$

Generally, if X has probability density :

$$f(x)=\begin{cases}\frac{1}{\theta} e^{-x/\theta} & x\geq 0\\ 0 & x<0\end{cases} \qquad \theta>0$$

Then X obeys to the **exponential distribution** with parameter θ.

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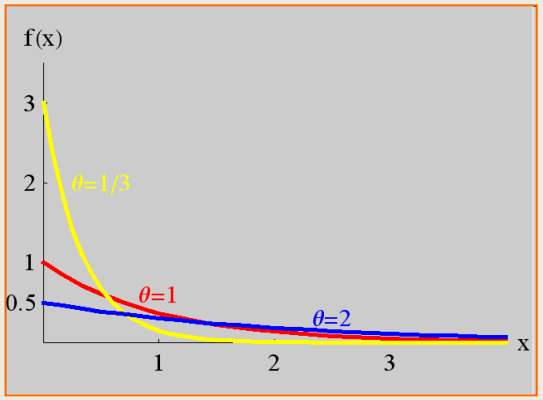
Exponential Distribution

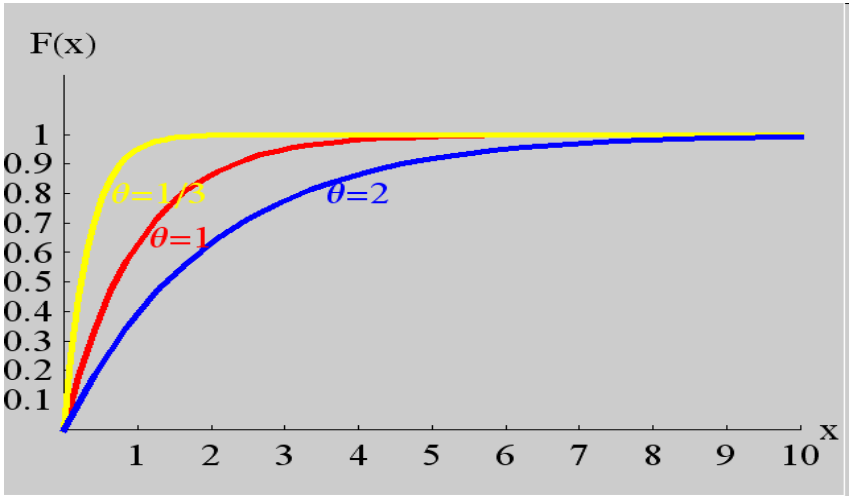
Definition 2.3.3: Let the PDF of continuous r.v. X be $f(x) = \begin{cases} \lambda e^{-\lambda x}, x>0 \\ 0, x \le 0 \end{cases}$,

in which $\lambda(>0)$ is a constant, then X has a **Exponential Distribution**, and marked as **X ~ Exp(λ)/X ~ E (λ)**.



Distribution function:

$$F(x) = \begin{cases} 1 - e^{-\lambda x}, x > 0, \\ 0, \qquad x \le 0. \end{cases}$$



Exponential Distribution

Application of Exponential Distribution:

The service life of some components or equipment follows a exponential distribution. For example, the life of radio components, the life of electrical equipment, the life of animals, etc., all have a exponential distribution.

Exponential Distribution

Example 2.3.3: Let the service life (in hours) of a certain type of daylight lamp X has a exponential distribution with λ=1/2000 .

Calculate:

(1) Take one lamp randomly, what is the probability of it can work more than 1000 hours.

(2) A lamp has worked 1000 hours, what the probability that it can work additional 1000 hours**?**

The distribution of common continuous random variables

Answer : the distribution function of X is

$$F(x)=\begin{cases}1-e^{-\frac{1}{2000}x}, & x\geq 0,\\ 0, & x<0.\end{cases}$$

$$\begin{aligned}(1)\ P\{X>1000\} &= 1-P\{X\leq 1000\}\\ &= 1-F(1000)\\ &= e^{-\frac{1}{2}}\approx 0.607.\end{aligned}$$

The distribution of common continuous random variables

Answer : the distribution function of X is

$$F(x)=\begin{cases}1-e^{-\frac{1}{2000}x}, & x\geq 0,\\ 0, & x<0.\end{cases}$$

$$(2)\ P\{X>2000|X>1000\}=\frac{P\{X>2000,X>1000\}}{P\{X>1000\}}$$

$$=\frac{P\{X>2000\}}{P\{X>1000\}}=\frac{1-P\{X\leq 2000\}}{1-P\{X\leq 1000\}}$$

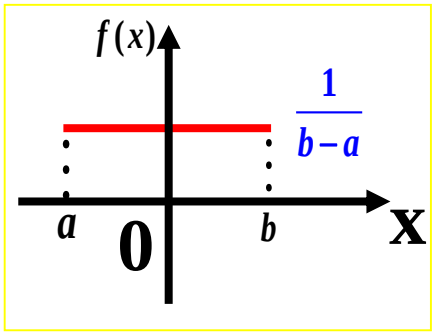
$$=\frac{1-F(2000)}{1-F(1000)}=e^{-\frac{1}{2}}\approx 0.607.$$

The distribution of common continuous random variables

Definition 2.3.2: Let the PDF of continuous r.v. X be

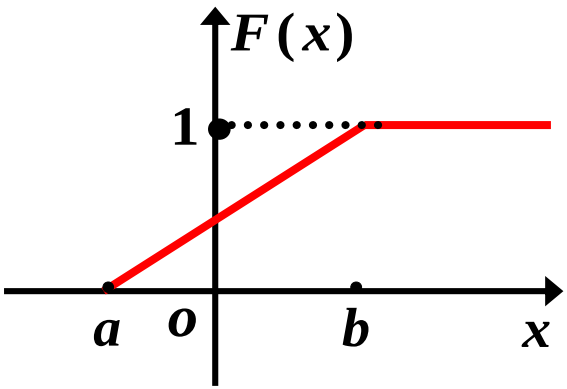
$$f(x)=\begin{cases}\frac{1}{b-a}, a<x<b\\ 0, Others\end{cases},$$

then X has a **uniform distribution** on interval (a, b), and marked as **X ~ U(a, b).**



Distribution function:

$$F(x)=\begin{cases}0, & x<a,\\ \frac{x-a}{b-a}, & a\le x<b,\\ 1, & x\ge b.\end{cases}$$



The distribution of common continuous random variables

Example 2.3.2: Suppose the random variable X has uniform distribution on [2, 5], then conduct three independent observations of X, calculate the probability of at least 2 observations of X is larger than 3.

Solution:

The probability density function of X is

$$f(x) = \begin{cases} \frac{1}{3}, 2 \le x \le 5 \\ 0, the\ other \end{cases}$$

A={the observations of X is larger than 3}

$$P(A) = P(X > 3) = \int_3^5 \frac{1}{3} dx = \frac{2}{3}$$

Let Y be the times that X is larger than 3 in 3 independent observations. $Y \sim B(3,2/3)$

$$P\{Y \ge 2\} = \binom{3}{2}\left(\frac{2}{3}\right)^2\left(1-\frac{2}{3}\right) + \binom{3}{3}\left(\frac{2}{3}\right)^3\left(1-\frac{2}{3}\right)^0 = \frac{20}{27}.$$

The distribution of common continuous random variables

Example. A bus station starts at 7 a.m. every 15 minutes, i.e. 7:00, 7:15, 7:30. If passengers arrive at the station, the time x is a uniform random variable between 7:00 and 7:30.

Calculate:

(1)Probability that the waiting time is **less than 5 minutes;**
(2)Probability that the waiting time **exceeds 10 minutes**.

The distribution of common continuous random variables

Answer:

Set 7:00 as the starting point 0, in minutes. That is, $X \sim U(0, 30)$

$$f(x) = \begin{cases} \dfrac{1}{30} & 0 < x < 30 \\ 0 & \text{else} \end{cases}$$

(1) $P\{10 < X < 15\} + P\{25 < X < 30\}$

$$= \int_{10}^{15} \frac{1}{30} dx + \int_{25}^{30} \frac{1}{30} dx = \frac{1}{3}$$

(2) $P\{0 < X < 5\} + P\{15 < X < 20\}$

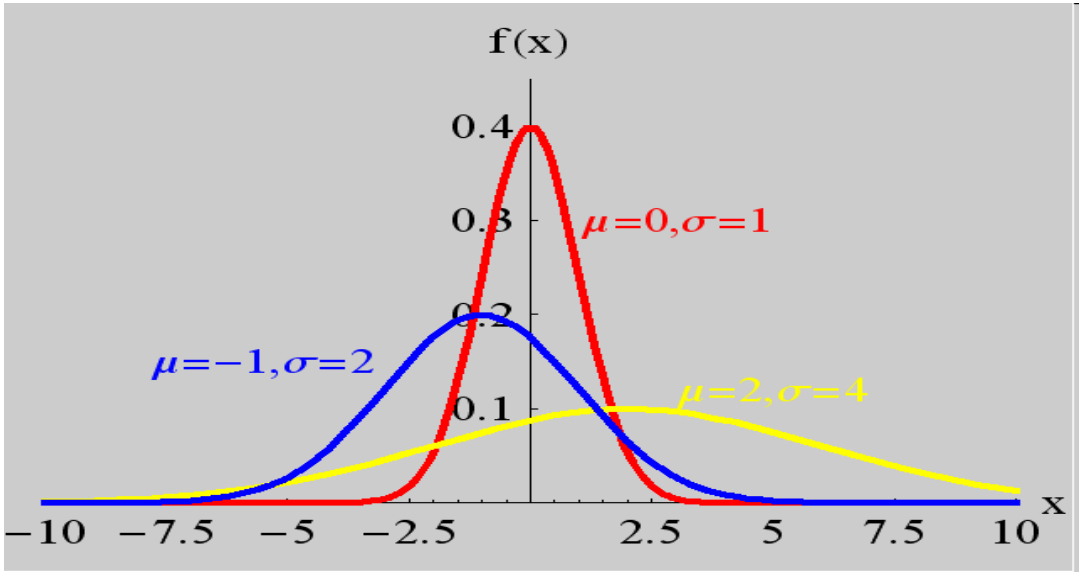
$$= \int_{0}^{5} \frac{1}{30} dx + \int_{15}^{20} \frac{1}{30} dx = \frac{1}{3}$$

The distribution of common continuous random variables

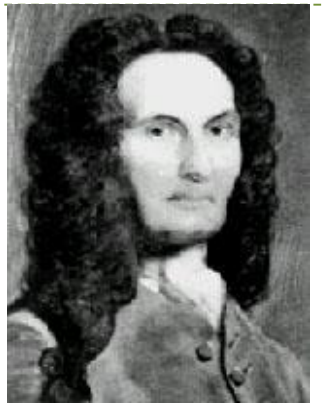
Definition 2.3.4: Let the PDF of continuous r.v. X be

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}, -\infty < x < +\infty$$

in which $\mu, \sigma (\sigma > 0)$ are constants, then X has a **Normal Distribution** or **Gaussian Distribution** with parameters μ, σ, and marked as **X ~ N(μ, σ^2).**



The distribution of common continuous random variables



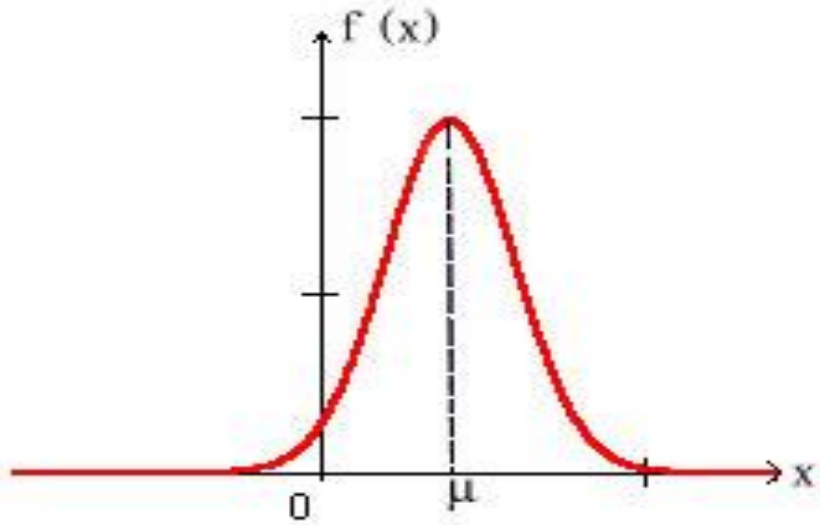
Abraham de Moivre

Gauss

- Normal distribution is the most widely used continuous distribution.
- Mathematician de moivre first discovered an approximate formula of binomial distribution, which is considered to be the first appearance of normal distribution.
- Normal distribution was popularized by mathematician Gauss before the 19th century, so it is usually called Gaussian distribution.

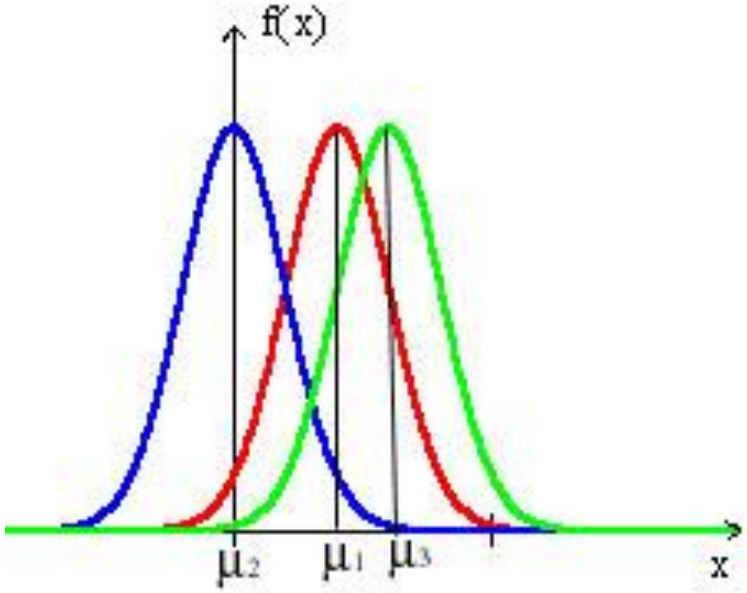
The distribution of common continuous random variables

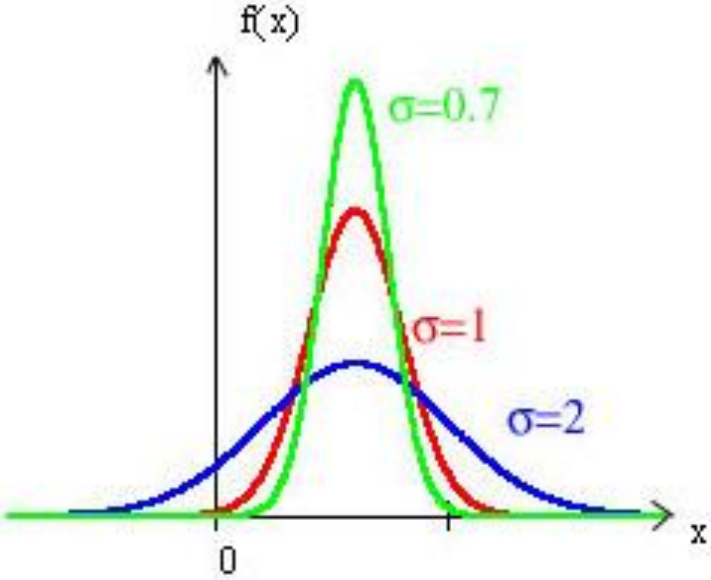
Graphic characteristics of normal distribution $N(\mu,\sigma^2)$



- The density curve of normal distribution is a bell curve about μ symmetry;
- It is characterized by “small at both ends, large in the middle and symmetrical left and right”.

The distribution of common continuous random variables



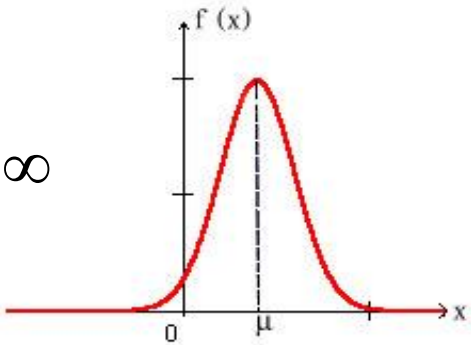


- μ determines the center position of the figure,
- σ determines the steepness of the peaks in the graph.
- when $x \to \pm\infty$, $f(x) \to 0$, x axis is a asymptotic line of the curve;

The distribution of common continuous random variables

Analyze the graphic characteristics of normal distribution by its pdf expression.

$$f(x)=\frac{1}{\sigma\sqrt{2\pi}}e^{-\frac{(x-\mu)^2}{2\sigma^2}},\quad -\infty<x<\infty$$



$f(x)\geq 0$

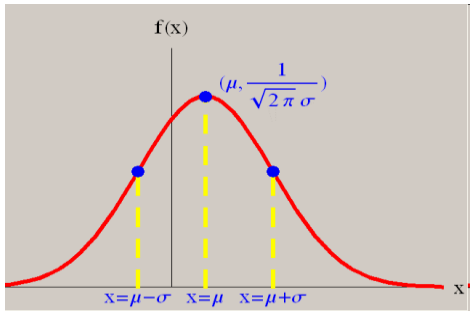
The entire probability density curve is above the x axis.

$f(x)$ symmetric at $x=\mu$, and reaches the maximum value

$$f(\mu)=\frac{1}{\sqrt{2\pi}\sigma} \text{ at } x=\mu .$$

The distribution of common continuous random variables

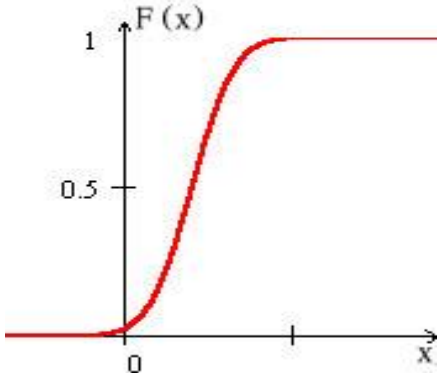
▲ $x = \mu \pm \sigma$ are two inflection points of the curve.



Distribution function of Normal distribution:

$$X \sim N(\mu, \sigma^2)$$

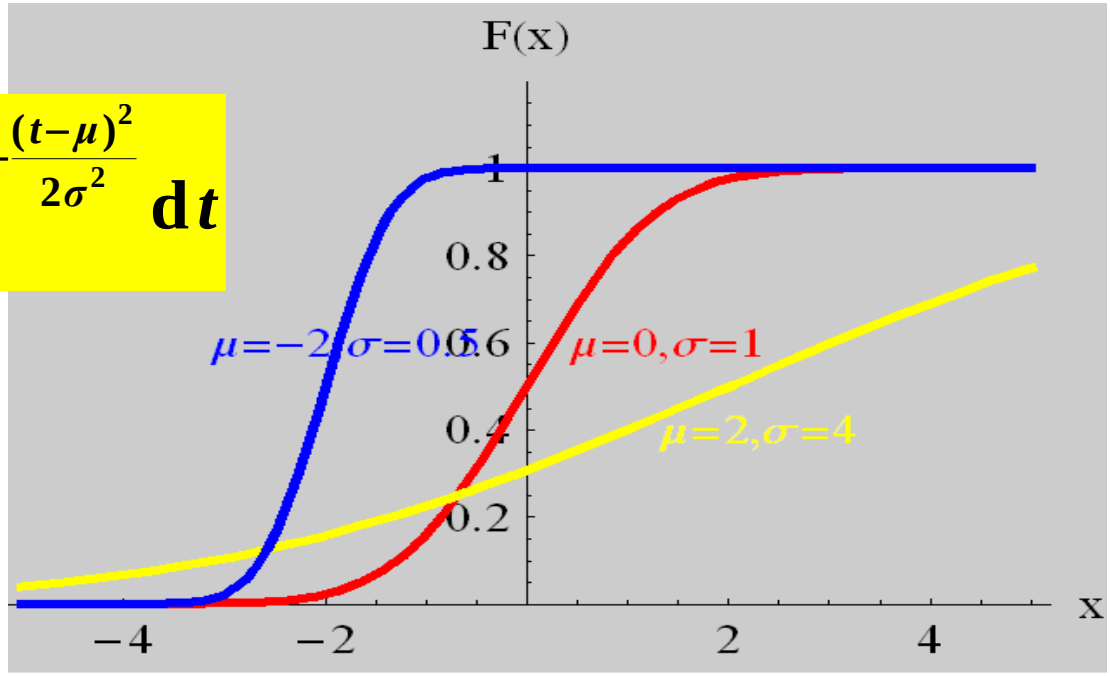
$$F(x) = \frac{1}{\sqrt{2\pi} \cdot \sigma} \int_{-\infty}^{x} e^{-\frac{(t-\mu)^2}{2\sigma^2}} dt, \quad -\infty < x < +\infty$$



The distribution of common continuous random variables

- The normal distribution is uniquely determined by its two parameters μ and σ.
- When parameters μ and σ change, they correspond to different normal distributions.

$$F(x)=\frac{1}{\sqrt{2\pi}\sigma}\int_{-\infty}^{x}e^{-\frac{(t-\mu)^2}{2\sigma^2}}\,\mathrm{d}t$$



The distribution of common continuous random variables

Geometrical feature of Normal distribution:

(1) x axis is a asymptotic line of the curve;

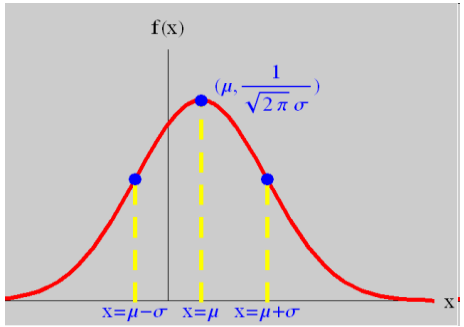
(2) The curve is symmetric about straight line $x = \mu$;

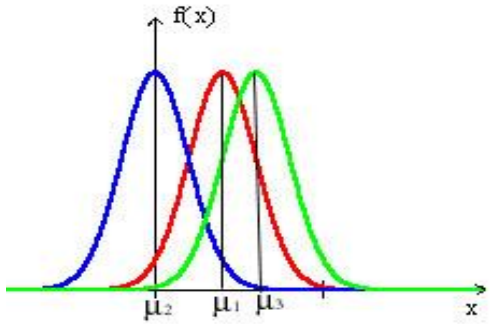
(3) When $x = \mu$, f(x) has a maximum value;

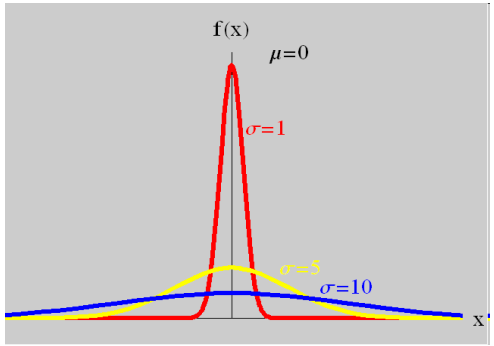
(4) $x = \mu \pm \sigma$ are two inflection points of the curve;

(5) If σ is fixed, let μ change, then f(x) is moving along x axis, and the shape of the graph doesn't change;

(6) If μ is fixed, let σ change, then f(x) will be flat when σ increase, and be steep while σ decrease.







The distribution of common continuous random variables

The background and application of Normal distribution:

Normal distribution is the most common and important distribution, measurement error, Human height and weight, the diameter, length, weight, height of the product and so on are approximately in normal distribution.

The distribution of common continuous random variables

The standard normal distribution:

When μ=0, σ=1, the normal distribution turns to the standard normal distribution, marked by $N(0,1)$. The PDF is

$$\varphi(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}, -\infty < x < +\infty$$

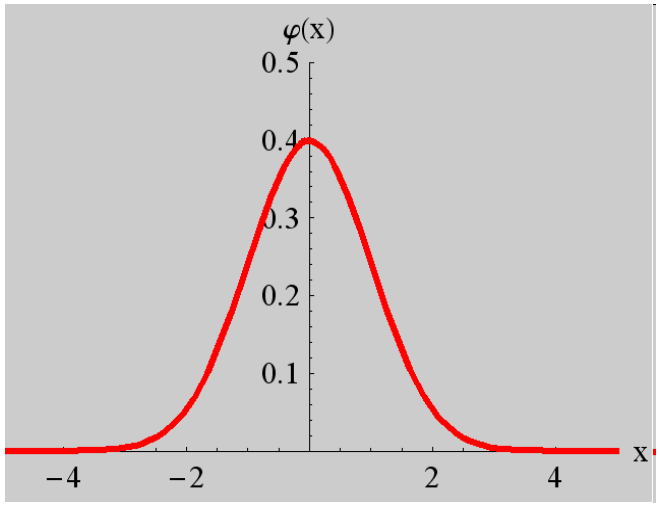
The distribution function is

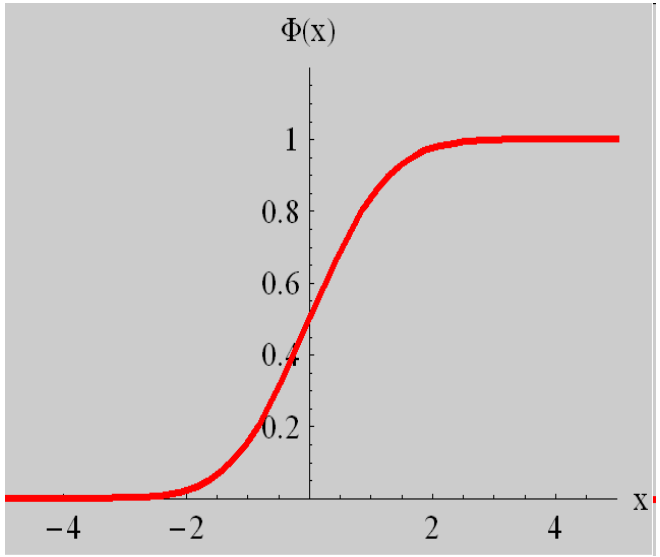
$$\Phi(x) = \int_{-\infty}^{x} \frac{1}{\sqrt{2\pi}} e^{-\frac{t^2}{2}} \mathrm{d}t, \quad -\infty < x < \infty.$$

The distribution of common continuous random variables

The standard Normal distribution:

$$\varphi(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}, \qquad -\infty < x < +\infty$$





$$\Phi(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{x} e^{-\frac{t^2}{2}} dt$$

The distribution of common continuous random variables

Example 2.3.5: X ~ N(0, 1), calculating P{1.25<X<2}

Solution:

$$P\{1.25 \le X < 2\} = \Phi(2) - \Phi(1.25)$$
$$= 0.9772 - 0.8944$$
$$= 0.0828.$$

Example 2.3.6: X ~ N(μ, σ^2), calculating P{c<X<d}

$$P\{c \le X \le d\} = \int_c^d \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}} \mathrm{d}x$$

Let $\frac{x-\mu}{\sigma} = u,$

$$= \int_{\frac{c-\mu}{\sigma}}^{\frac{d-\mu}{\sigma}} \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{u^2}{2}} \cdot \sigma \mathrm{d}u = \int_{\frac{c-\mu}{\sigma}}^{\frac{d-\mu}{\sigma}} \frac{1}{\sqrt{2\pi}} e^{-\frac{u^2}{2}} \cdot \mathrm{d}u$$

$$= \int_{-\infty}^{\frac{d-\mu}{\sigma}} \frac{1}{\sqrt{2\pi}} e^{-\frac{u^2}{2}} \cdot \mathrm{d}u - \int_{-\infty}^{\frac{c-\mu}{\sigma}} \frac{1}{\sqrt{2\pi}} e^{-\frac{u^2}{2}} \cdot \mathrm{d}u$$

$$= \Phi\left(\frac{d-\mu}{\sigma}\right) - \Phi\left(\frac{c-\mu}{\sigma}\right).$$

The distribution of common continuous random variables

▲ Importance of standard normal distribution

Any general normal distribution can be transformed into standard normal distribution by linear transformation $\frac{x-\mu}{\sigma}$.

Lemma: *If* $X \sim N(\mu, \sigma^2)$, then $Z = \frac{X-\mu}{\sigma} \sim N(0,1)$.

Proof: $$P\{Z \le x\} = P\left\{\frac{X-\mu}{\sigma} \le x\right\} = P\{X \le \mu + \sigma x\}$$

$$= \frac{1}{\sqrt{2\pi}\sigma}\int_{-\infty}^{\mu+\sigma x} e^{-\frac{(t-\mu)^2}{2\sigma^2}}\,\mathrm{d}t,$$

Let $\frac{t-\mu}{\sigma} = u,$

$$P\{Z \le x\} = \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{x} e^{-\frac{u^2}{2}}\,\mathrm{d}u = \Phi(x),$$

$$\Rightarrow \quad Z = \frac{X-\mu}{\sigma} \sim N(0,1).$$

The distribution of common continuous random variables

▲ Importance of standard normal distribution

Any general normal distribution can be transformed into standard normal distribution by linear transformation $\frac{x-\mu}{\sigma}$.

$$Z = \frac{X-\mu}{\sigma} \sim N(0,1).$$

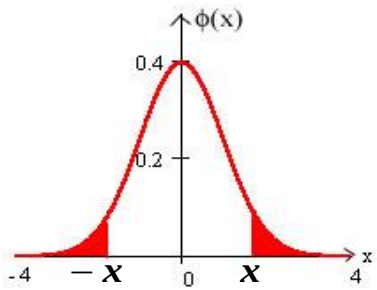
▲ Thus , *if* $X \sim N(\mu, \sigma^2)$, then its CDF as:

$$F(x) = P(X \le x) = P\left(\frac{X-\mu}{\sigma} \le \frac{x-\mu}{\sigma}\right)$$

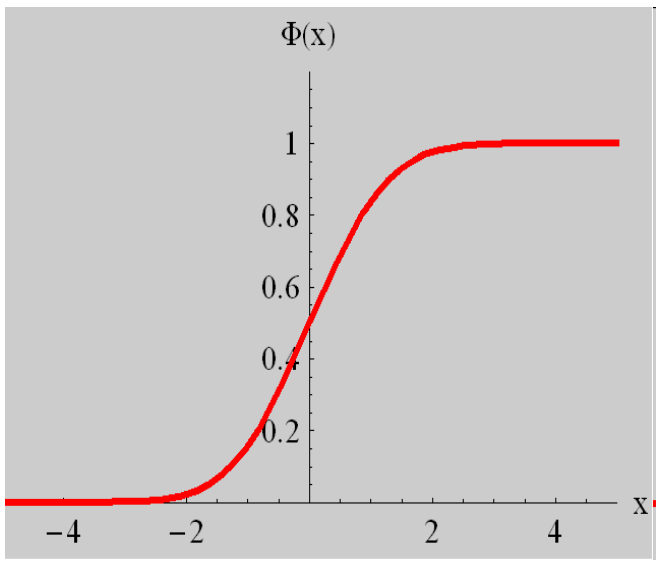
$$= \Phi\left(\frac{x-\mu}{\sigma}\right)$$

The distribution of common continuous random variables

▲ Standard normal distribution table



$$\Phi(x)=\frac{1}{\sqrt{2\pi}}\int_{-\infty}^{x}e^{-\frac{t^2}{2}}dt$$



(1) When $x>0$, the value of Φ(x) is given in the table.

(2) When $-x<0$:

$$\Phi(-x)=1-\Phi(x)$$

The distribution of common continuous random variables

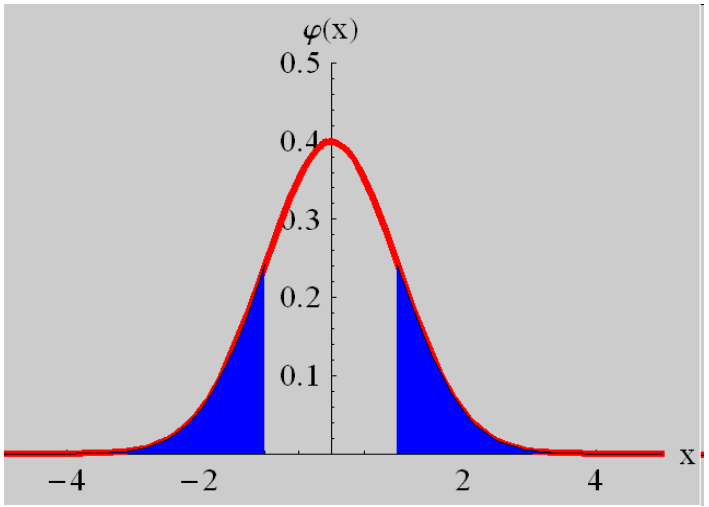
Example 2.3.7: Proving $\Phi(-x)=1-\Phi(x)$.

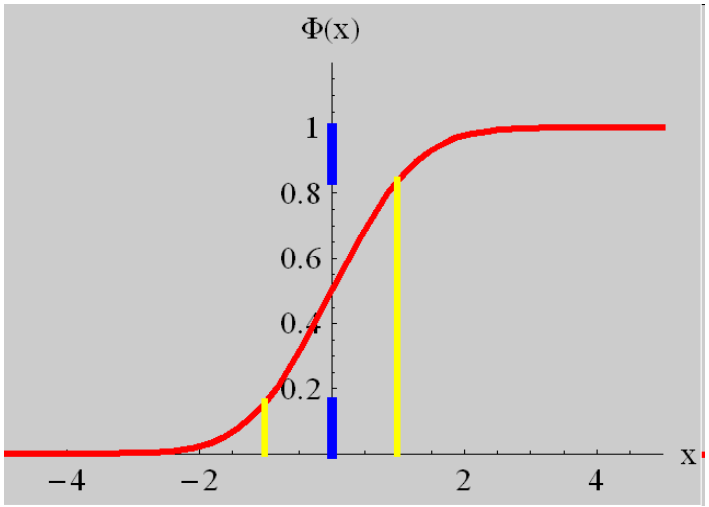
Proof:

$$\Phi(-x)=\int_{-\infty}^{-x}\frac{1}{\sqrt{2\pi}}e^{-\frac{x^2}{2}}\,\mathrm{d}x$$

$$x=-x \quad \Rightarrow \quad =\int_{x}^{\infty}\frac{1}{\sqrt{2\pi}}e^{-\frac{x^2}{2}}\,\mathrm{d}x=\int_{-\infty}^{\infty}\frac{1}{\sqrt{2\pi}}e^{-\frac{x^2}{2}}\,\mathrm{d}x-\int_{-\infty}^{x}\frac{1}{\sqrt{2\pi}}e^{-\frac{x^2}{2}}\,\mathrm{d}x$$

$$=1-\Phi(x).$$





The distribution of common continuous random variables

- *if* $X \sim N(0,1)$, then:

$$P(a < X < b) = \Phi(b) - \Phi(a)$$

- *If* $X \sim N(\mu, \sigma^2)$, $Y = \dfrac{X-\mu}{\sigma} \sim N(0,1)$

then: $$P(a < X < b) = P\left(\frac{a-\mu}{\sigma} \le Y \le \frac{b-\mu}{\sigma}\right)$$

$$= \Phi\left(\frac{b-\mu}{\sigma}\right) - \Phi\left(\frac{a-\mu}{\sigma}\right)$$

For any interval $(x_1, x_2]$,

$$P(x_1 < X \le x_2) = P\left(\frac{x_1-\mu}{\sigma} < \frac{X-\mu}{\sigma} \le \frac{x_2-\mu}{\sigma}\right)$$

$$= \Phi\left(\frac{x_2-\mu}{\sigma}\right) - \Phi\left(\frac{x_1-\mu}{\sigma}\right)$$

The distribution of common continuous random variables

3 σ Rule

It is calculated by looking up the distribution function table of standard normal distribution.

The following can be obtained: When $X \sim N(0, 1)$:

$$P(|X| \leq 1) = 2\Phi(1) - 1 = 0.6826$$

$$P(|X| \leq 2) = 2\Phi(2) - 1 = 0.9544$$

$$P(|X| \leq 3) = 2\Phi(3) - 1 = 0.9974$$

Its shows that the value of X is almost concentrated in [- 3,3], and the possibility of exceeding this range is less than 0.3%.

The distribution of common continuous random variables

The above conclusions can extended to the general normal distribution, when $Y \sim N(\mu, \sigma^2)$

$$P(|Y-\mu| \leq \sigma) = 0.6826$$

$$P(|Y-\mu| \leq 2\sigma) = 0.9544$$

$$P(|Y-\mu| \leq 3\sigma) = 0.9974$$

Almost all values of Y are concentrated in interval [μ±3σ].

This is statistically referred to as the 3σ Rule.

(Triple standard deviation principle)

Section 2.3 Continuous random variable and its probability density

- Probability density and its properties
- Cumulative distribution function of continuous random variable
- The distribution of common continuous random variables
- Examples

The distribution of common continuous random variables

Example: It is assumed that the length x (mm) of the parts produced by the lathe follows the normal distribution $N(50, 0.75^2)$, the products are qualified if it length is between 50 ± 1.5mm.

Calculate: the probability that the production parts are qualified products.

Answer:

$$\because \quad X \sim N(50, 0.75^2)$$

$$\therefore \quad P(|X-50|<1.5) = P(48.5 < X < 51.5)$$

$$= \Phi\left(\frac{51.5-50}{0.75}\right) - \Phi\left(\frac{48.5-50}{0.75}\right)$$

$$= \Phi(2) - \Phi(-2)$$

$$= \Phi(2) - (1-\Phi(2)) = 2\Phi(2) - 1$$

$$= 2 \times 0.9772 - 1 = 0.9544$$

The distribution of common continuous random variables

Example. Place a temperature regulator in a container containing a liquid. When the regulator is adjusted at d ℃, the temperature X of the liquid (in ℃) is a random variable,
and $X \sim N(d, 0.5^2)$.

Calculate :

(1) If d = 90, find the probability that x is less than 89;

(2) If the probability of keeping the liquid temperature at least 80 is not less than 0.99, what is the minimum d?

Answer: (1) $P(X < 89) = P\left(\frac{X-90}{0.5} < \frac{89-90}{0.5}\right)$

$$= \Phi(-2) = 1 - \Phi(2)$$

$$= 1 - 0.9772 = 0.0228$$

The distribution of common continuous random variables

(2). According to the question, d should satisfy:

$$0.99 \le P(X > 80) = P\left(\frac{X-d}{0.5} > \frac{80-d}{0.5}\right)$$

$$= 1 - P\left(\frac{X-d}{0.5} \le \frac{80-d}{0.5}\right) = 1 - \Phi\left(\frac{80-d}{0.5}\right)$$

$$\Phi\left(\frac{80-d}{0.5}\right) \le 1 - 0.99 = 0.01$$

What should we do?

$\Phi(x)$ =0.01 cannot be found in Standard normal distribution table.

BUT!

The distribution of common continuous random variables

$\Phi(x)$ =0.01 cannot be found in Standard normal distribution table.

We can do like that:

$$\because \quad \Phi(-u) = 1 - \Phi(u) \qquad \therefore \quad \Phi(u) = 1 - \Phi(-u)$$

Let $1 - \Phi(-u) \le 0.01 \quad \Rightarrow \quad \Phi(-u) \ge 0.99$

Find in Schedule, $-u \ge 2.32$

That is , $\dfrac{80 - d}{0.5} \le -2.32$

Thus ,we find $d > 81.16$

The distribution of common continuous random variables

Example. The height of the subway door in a city is designed according to the chance that the man meets the top of the door below 0.01. It is assumed that the man's height obeys $X \sim N(170, 6^2)$.

Question: how should the door height be determined.

Answer:

Set the door height as *h* (cm).

According to the design requirements, we should find the minimum *h* that can satisfy $\mathbf{P(x \geq h) \leq 0.01}$

or $\mathbf{P(x < h) \geq 0.99}$.

The distribution of common continuous random variables

$$\because \ X \sim N(170, 6^2), \quad \therefore \ \frac{X-170}{6} \sim N(0,1)$$

$$\text{so,} \quad P(X<h) = \Phi\left(\frac{h-170}{6}\right) \geq 0.99$$

Find in Schedule, $\Phi(2.33) = 0.9901 > 0.99$

$$\text{So,} \quad \frac{h-170}{6} = 2.33$$

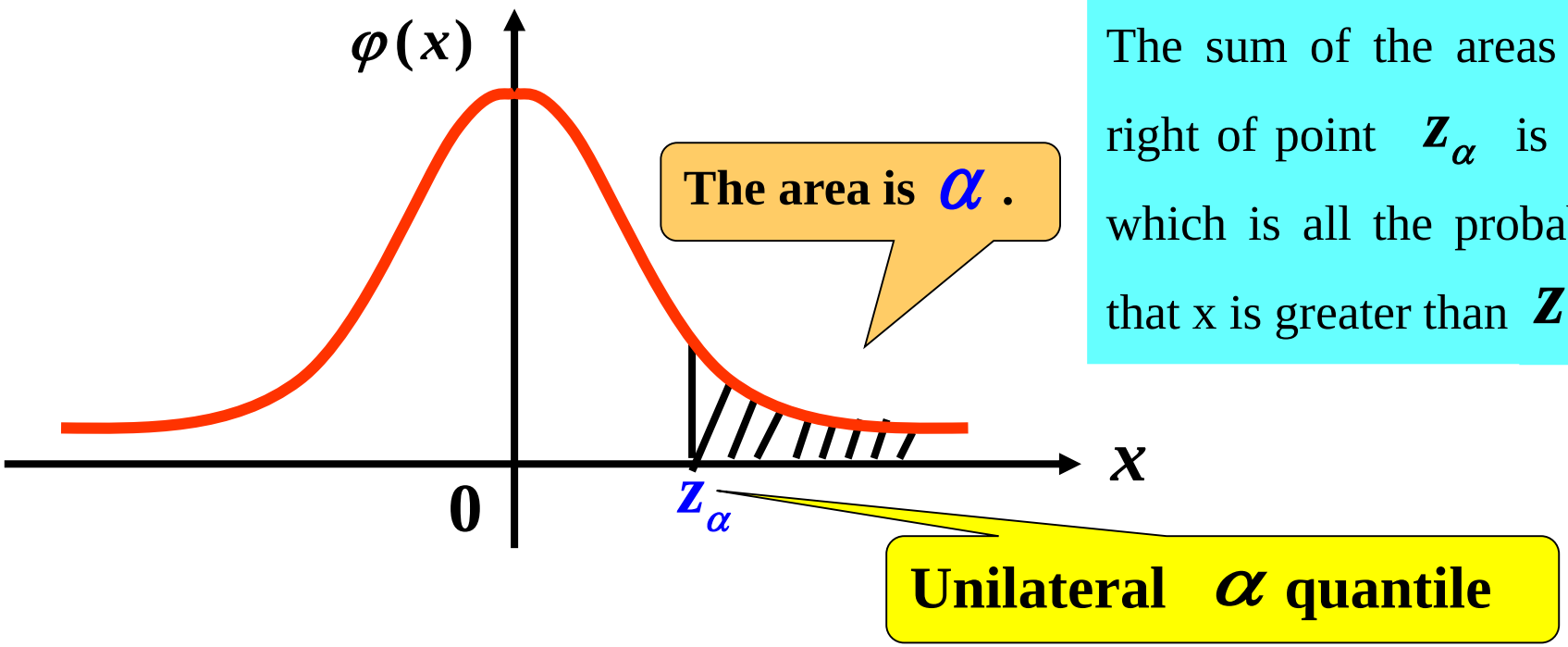
$$\text{Thus}, \ h = 170 + 13.98 \approx 184$$

Conclusion: when the design height of the door is 184cm, the chance of men meeting the door does not exceed 0.01.

The distribution of common continuous random variables

Concept about α Quantile

Definition: $X \sim N(0,1)$, if Z_α satisfies $P(X > z_\alpha) = \alpha$, then, Z_α is said to be the α quantile on the standard normal distribution.



The sum of the areas to the right of point z_α is α, which is all the probabilities that x is greater than z_α .

The distribution of common continuous random variables

▲ $\Phi(z_\alpha) = \int_{-\infty}^{z_\alpha} f(x)dx = 1-\alpha$

$=$ Subtract the area to the right of point z_α from the whole area

$=$ The area from $-\infty$ to z_α available in Schedule 2

▲ How to find the value of z_α from the normal distribution table:

For a given α, then:

$z_\alpha = z$ value corresponding to probability $(1-\alpha)$.

Eg : $z_{0.05} = z(1-0.05) = z(0.95) = 1.645$

It can be verified that:

$$\Phi(1.645) = 1-\alpha = 1-0.05 = 0.95$$

The distribution of common continuous random variables

Or ,

$$z_{0.005} = z(1-0.005) = z(0.995) = 2.57$$

$$\Phi(2.57) = 1 - 0.005 = 0.995$$

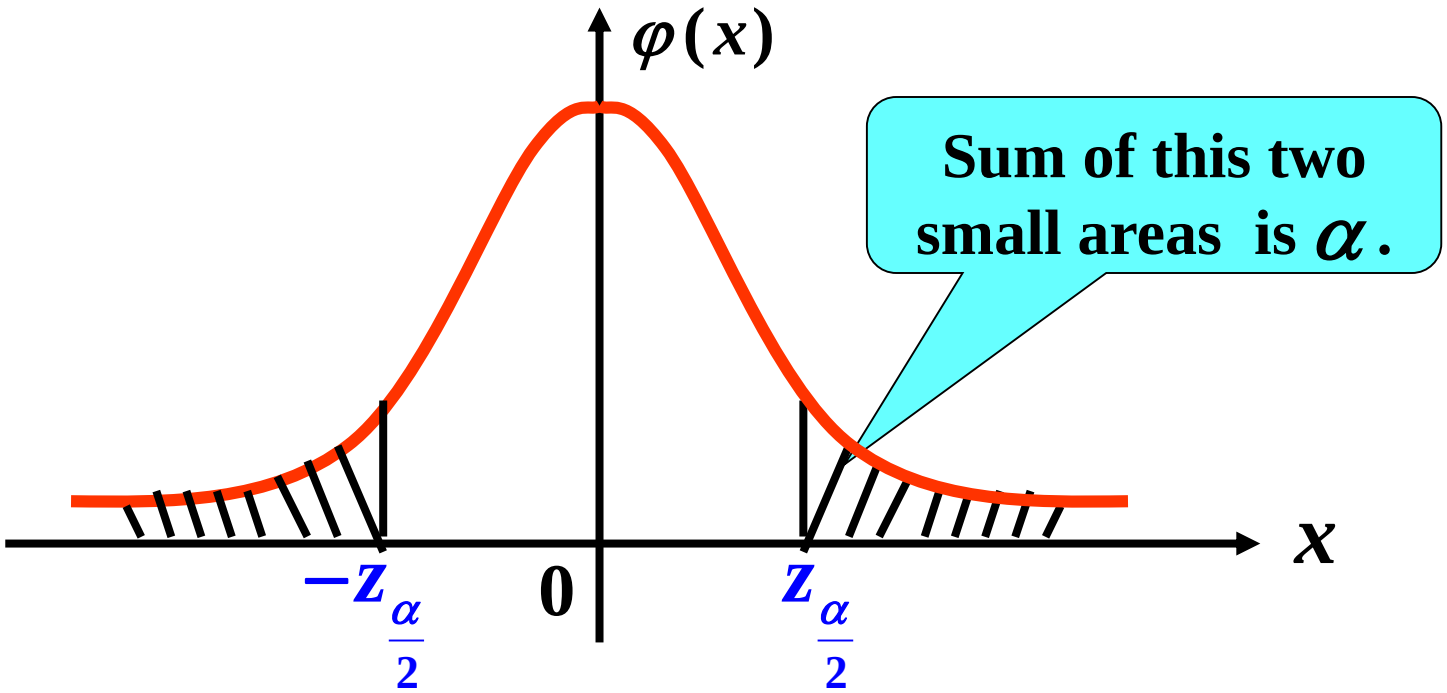
$$z_{0.001} = z(1-0.001) = z(0.999) = 3.10$$

$$\Phi(3.10) = 1 - 0.001 = 0.999$$

The distribution of common continuous random variables

Definition of bilateral α quantile

If $P(|X| > Z_{\alpha/2}) = \alpha$, then $Z_{\alpha/2}$ is called the **bilateral α quantile** on the standard normal distribution.



The distribution of common continuous random variables

Homeworks:

5.6.1 / 5.6.2 / 5.6.5 / 5.6.7 / 5.6.8